

Sorting

* Selection Sort :-

Pehle smallest element of select karo phir phir usko first element ke saath swap kar do, phir 2nd smallest dhundho or usko 2nd element ke saath swap karo do, or ye process of repeat karo.

Ex:

9	7	3	1	6
---	---	---	---	---

 $n = 5 \text{ element}$
↓ ↑
smallest largest

Round 1 $\rightarrow$ 1 7 3 9 6

Round 2 $\rightarrow$ 1 3 7 9 6

no. of round $y = (n-1)$ round

Round 3 $\rightarrow$ 1 3 6 9 7

Round 4 $\rightarrow$ 1 3 6 7 9

sorted

Ex:

10	8	2	3	4
----	---	---	---	---

 $n = 6 \text{ element}$
↓ ↑
smallest largest

(0-5) Round 1 $\rightarrow$ 1 8 2 3 10 4

(1-5) 2 $\rightarrow$ 1 2 8 3 10 4

(2-5) 3 $\rightarrow$ 1 2 3 8 10 4

(3-5) 4 $\rightarrow$ 1 2 3 4 10 8

(4-5) 5 $\rightarrow$ 1 2 3 4 8 10

no. of round 5, or

$(n-1)$ round

Swap
Karna
hai

```
for (i=0; i < n-1; i++) {
```

```
    index = i;
```

```
    for (j=i+1; j < n; j++) {
```

```
        if (arr[j] < arr[index]) {
```

```
            index = j;
```

```
        }
```

```
        swap(arr[index], arr[i]);
```

```
    }
```

```
}
```

```
print arr.
```